

26th Experiment

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.metrics import accuracy_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.utils import to_categorical

# Generate Multi-Class Dataset
X, y = make_blobs(
    n_samples=500,
    centers=3,
    n_features=2,
    cluster_std=1.5,
    random_state=42
)

# Convert labels to One-Hot Encoding
y_cat = to_categorical(y)

# Build Neural Network
model = Sequential([
    Dense(16, activation='tanh', input_shape=(2,)),
    Dense(8, activation='tanh'),
    Dense(3, activation='softmax')
])

# Compile Model
```

```
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# Train Model
history = model.fit(
    X,
    y_cat,
    epochs=50,
    verbose=0
)

# Training Loss Graph
plt.figure(figsize=(5,3))
plt.plot(history.history['loss'])
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(True)
plt.show()

# Prediction
pred = model.predict(X)
pred_class = np.argmax(pred, axis=1)

# Accuracy
acc = accuracy_score(y, pred_class)
print("Accuracy:", round(acc*100,2), "%")
```

```
# Dataset Visualization
```

```
plt.figure(figsize=(5,4))
```

```
plt.scatter(X[:,0], X[:,1], c=y)
```

```
plt.title("Multi-Class Data (Tanh Activation)")
```

```
plt.xlabel("Feature 1")
```

```
plt.ylabel("Feature 2")
```

```
plt.show()
```

Output:

